

IFS 353

DOCUMENTATION AND REPORTING

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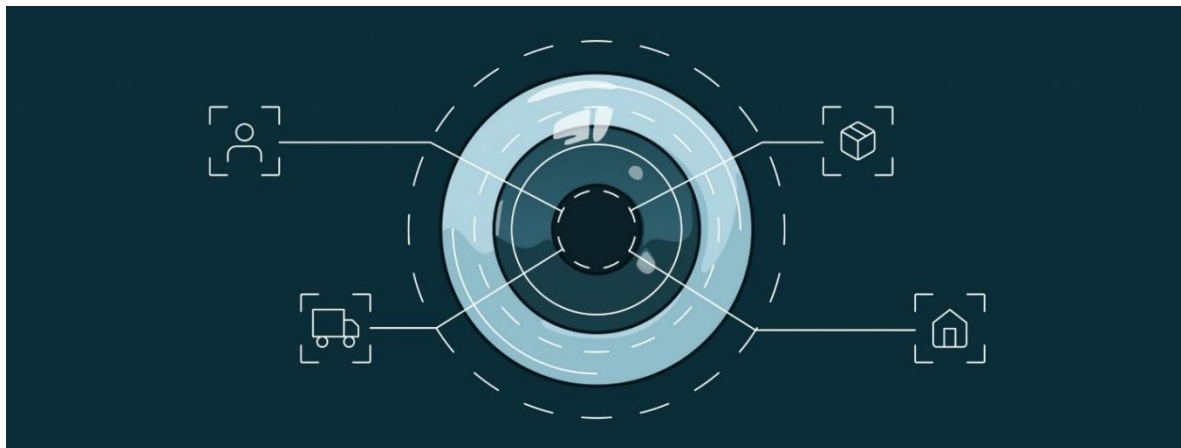


INTRODUCTION

This report and documentation give a brief overview into a machine learning based vision system I have developed that aims at solving a real-world classification problem. My model is based on distinguishing between different IoT sensors, and this report will be a guide to the development of the model. Topics will range from the collection of data, model development and training, performance and taking a deep dive into ethical considerations of deploying such a system into the real world.

A potential real-world application of the system is in maintenance and defects. The model can be used to identify defects, wear and tear over time or real time on equipment before they lead to failure or poor performance thus reducing reliance on manual inspections. In a greenhouse environment with CO, CO₂, DHT22/11, motion sensors machine learning image classification can be used to take images of these components in both normal and defective states and the system over time could classify each component as defective or functional based on corrosion, cracks or wear and tear.

There are several limitations to the application of this system to a real-world problem above. The first one is the quantity and quality of data. For the model to be perfectly accurate and perform consistently, a significant amount of data would be required to train and develop the model. There needs to be sufficient data on equipment that is normal and defective that the model can train on. Another limitation is environmental factors, factors such as lighting in a greenhouse environment can affect the ability of the model from accurately identifying defects. Other limitations include real time processing, if this is a requirement, it could require sufficient computational resources. Maintenance and model updating and training could be a limitation, in greenhouses conditions change inconsistently and the model would need to be updated regularly to remain accurate as conditions change and have an effect on the quality of the equipment.



DATA COLLECTION AND PRE-PROCESSING

1.1. Data collection

❖ Description:

- The source of the dataset/images for the image classification machine learning model were obtained through online repositories such as Google images and captured images by a mobile cellphone. The purpose was to provide the model with different types of quality dataset/images to help train and enhance the quality of the model.

❖ Classes/objects:

- There are 6 different classes or objects used for modeling and training and these are IoT devices or components namely smart watches, carbon dioxide sensors, raspberryPI's, smart dash cameras, dht11 sensor and motion sensors.

❖ For the number of images obtained for each class:

- Carbon dioxide sensors: 26 images(.JPG).
- Motion sensors: 126 images(.JPG, .PNG, .JPEG).
- Raspberry PI: 173 images(.JPG, .PNG, .JPEG).
- Smart dash camera: 183 images(.JPG, .PNG, .JPEG).
- Smart watches: 152 images(.JPG, .PNG, .JPEG).
- DHT11 sensor: 26 images(.JPG).

1.2. Data Pre-processing

❖ Image resizing :

- All images were preprocessed and resized to a 64x64. Resizing the images was important because it could help reduce the file size of some images, reduce data usage and potentially help improve visual quality. This was also done to have a uniform size for all of the images.

❖ Normalization :

- Normalization was important in order to help the model train more effectively and the goal of normalization is to transform features to be on a similar scale.
[X = X.astype('float32') / 255.0] In my case normalizing the image data involved converting the pixel value of the images into a floating-point format where the values lie between 0 and 1.
Image normalization ensures optimal comparisons across data acquisition methods as well as speeding the learning process.

1.3. Data Splitting

- ❖ Training, validation and test sets:

- The training, validation, and test split is fundamental to the development of the machine learning model because the model should respond well to new, unseen data and not only rely on test data.

The training set data is reserved to fit the model, and the model sees and learns from this set and continuously improves its parameters based on it.

The validation set is used for refinement, evaluation and fine tuning the model during training which helps assess performance and adjust parameters.

The test data is simply the dataset that is used to evaluate the final performance of the model. (See Appendix A - Figure 1)

- ❖ Rationale for splitting ratio:

- The rough standard for train-validation-test splits is 60-80% training data, 10-20% validation data, and 10-20% test data and I used middle value for each split such as 70% training data, 15% validation data and 15% testing data.

MODEL DEVELOPMENT

2.1. Model of choice:

- ❖ In addition to developing my own convolutional neural network model I found it beneficial to add another CNN model in the form of VGG16. This CNN is considered one of the best computer vision models to date being able to classify 1000 images of 1000 different categories with 92.7% accuracy. It is regarded as one of the popular algorithms for image classification and is easy to use with transfer learning.

2.2. Framework of choice:

- ❖ The framework of choice was mostly influenced by tutorials I watched on YouTube and GitHub repositories on object detection and image classification I found on GitHub. But with more use it proved to be the right choice as I found it relatively easy to use and some of its libraries were easy to install as well as having detailed documentation that explains the framework in depth and how best to utilize it. Tensorflow houses several attributes such as scalability, versatile API and TensorBoard which is a unique tool for visualization.

2.3. Improving model performance:

- ❖ Machine learning models rely heavily on enormous amounts of data it was my priority to get as much data as possible as well as having different variations of the data such as using images from the internet and data collection repositories, images taken from my mobile phone in different settings and environments. This is beneficial in helping my model make accurate results.
- ❖ The last two techniques I utilized which have been mentioned above were data augmentation and transfer learning. Data augmentation is a popular technique that is widely adopted especially in vision systems where the model relies on real world data that may prove difficult to capture. And what it does is that It retrieves the current dataset and continuously modifies it to create new variations of the images. It would take one image, resize, flip, change the colour a bit to improve the model performance.
- ❖ Transfer learning was used in the form of pre-trained models such as VGG16 which is a model that has been trained with enormous datasets before and is a popular algorithm for image classification

MODEL EVALUATION

3.1. This section will highlight and evaluate the performance of the image classification model using multiple metrics which include calculating the total accuracy, plotting the validation and test data accuracy and loss, accuracy, precision, recall, F1 score and lastly confusion matrix. These metrics provide insights into the model's ability to classify images of different IoT sensors. I will highlight what each of these metrics represent in terms of the model's performance (See Appendix A – Figure 2, 3, 4):

- ❖ Test accuracy:
 - ❖ The test accuracy shows how often a classification machine learning model is correct overall and my model gave a 93% test accuracy.
- ❖ Precision:
 - ❖ Precision shows how often a machine learning model is correct when predicting the targeted class and for the motion sensor the model achieved a 1.00 precision followed by the CO2 sensor a dht11 sensor at 0.95 and 0.94 respectively and lastly 0.88 for the raspberry pi.
- ❖ Recall:
 - ❖ Recall simply shows whether the machine model can find all of the objects of the target class and my model achieved 1.00 for the raspberry pi and CO2 sensor and 0.79 and 0.94 for the motion sensor and dht11 respectively.
- ❖ F1-score:
 - ❖ The F1-score measures the mean between the recall and precision scores and an F1 score above 0.9 is considered excellent which the model was able to achieve for the raspberry pi, CO2 sensor, dht11 and a F1 score between 0.8 and 0.9 is considered good which is what the motion sensor achieved. (See Appendix B – Figure 5)

3.2. Strengths and limitations:

- ❖ Strengths:

The models excel in identifying the raspberry pi and carbon dioxide sensor and the addition of the pre trained model helped strengthen the performance of the model. By referencing the confusion matrix, we can assess the model is performing extremely good in classifying the raspberry pi and carbon dioxide.

```
[15 2 1 1] Motion sensor
[ 0 23 0 0] raspberry pi
[ 0 0 19 0] co2 sensor
[ 0 1 0 17] dht11
```

❖ Limitations:

The model struggles to properly classify the dht11 and motion sensor and this is also highlighted in the confusion matrix where there are off diagonal values which represent incorrect predictions in the models, and this is where the model predicts one thing but labels a different value.

```
[15 2 1 1] Motion sensor
[ 0 23 0 0] raspberry pi
[ 0 0 19 0] co2 sensor
[ 0 1 0 17] dht11
```

ETHICAL CONSIDERATIONS

❖ Potential Biases:

The biggest ethical concerns rising in AI and ML models are fairness and bias. Working with data that contains biases can cause algorithms to perpetuate and even exacerbate those biases. Potential biases in my dataset include an uneven number of images for all the classes and therefore some classes could have the model performing poorly on them. The model struggles classifying the dht11 sensor and motion sensor, they are often misclassified as other classes and simply not detected by the model or there is a very long responsive rate from the model to classifying the component as the correct value.

Measures such as data augmentation and having images from both the internet and phone camera in different settings were sourced to reduce bias.

❖ Proposals for responsible use and privacy concerns:

- Data privacy and security: Data privacy is another big ethical concern in machine learning. Machine learning models rely heavily on data and one challenge with sourcing this data is user privacy and personal data. ensure that any sensitive data is encrypted where necessary and data is collected ethically.
- Ethical AI and fairness: Implement methods to remove biases in the model's predictions such as auditing the model.
- Transparency: The model should be able to explain to the user why a certain equipment is defective, this increases trust in the model's effectiveness and leads to quicker human intervention.
- Retrain the model regularly: The model should be continuously retrained and updated with new data of new types of wear and tear on equipment.

APPENDIX A

Figure 1

Training set shape: (366, 224, 224, 3) (366,)
Validation set shape: (78, 224, 224, 3) (78,)
Test set shape: (79, 224, 224, 3) (79,)
Shape of X: (523, 224, 224, 3)
Shape of y: (523,)

Figure 2

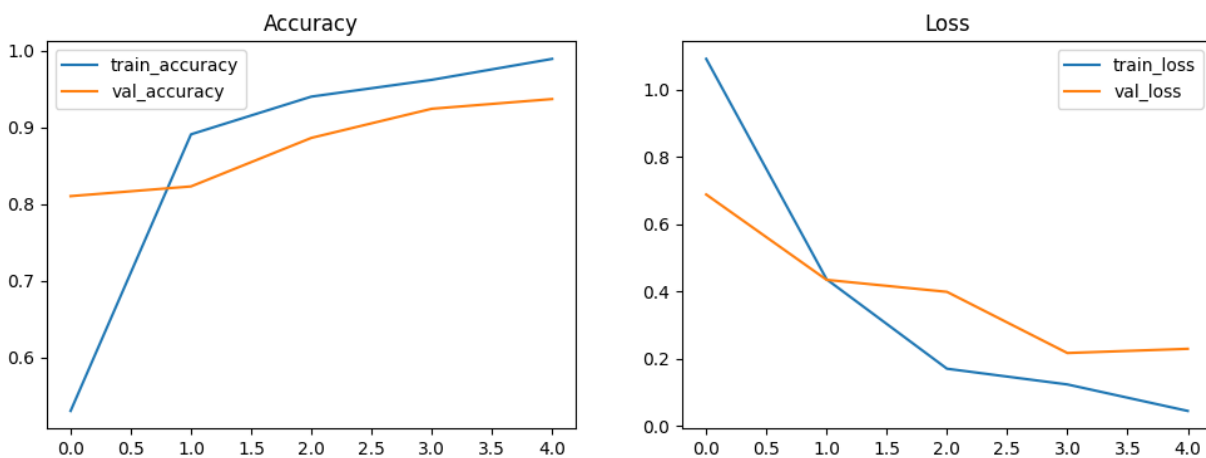


Figure 3

Epoch 1/5
12/12 ————— 274s 22s/step - accuracy: 0.4211 - loss: 1.2210 - val_accuracy: 0.8101 - val_loss: 0.6876
Epoch 2/5
12/12 ————— 355s 25s/step - accuracy: 0.8826 - loss: 0.5096 - val_accuracy: 0.8228 - val_loss: 0.4338
Epoch 3/5
12/12 ————— 270s 23s/step - accuracy: 0.9399 - loss: 0.1851 - val_accuracy: 0.8861 - val_loss: 0.3981
Epoch 4/5
12/12 ————— 281s 24s/step - accuracy: 0.9546 - loss: 0.1269 - val_accuracy: 0.9241 - val_loss: 0.2162
Epoch 5/5
12/12 ————— 281s 23s/step - accuracy: 0.9931 - loss: 0.0414 - val_accuracy: 0.9367 - val_loss: 0.2287

Figure 4

3/3 ————— 64s 18s/step - accuracy: 0.9176 - loss: 0.2650
Test accuracy: 0.9367088675498962

APPENDIX B

Figure 5

```
Confusion Matrix:
[[15  2  1  1]
 [ 0 23  0  0]
 [ 0  0 19  0]
 [ 0  1  0 17]]
Classification Report:
              precision    recall  f1-score   support

motionsensor      1.00      0.79      0.88        19
  raspberrypi      0.88      1.00      0.94        23
    cosensor      0.95      1.00      0.97        19
      dht11      0.94      0.94      0.94        18

 accuracy              0.94              79
 macro avg      0.94      0.93      0.93        79
weighted avg      0.94      0.94      0.94        79
```

APPENDIX C

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